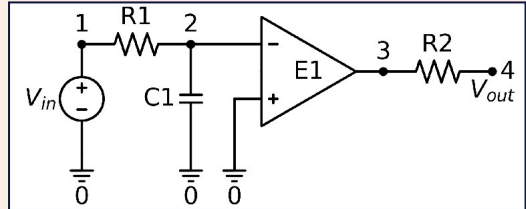




Circuit Symbolic Expression Generation

Question: Based on the circuit diagram, please calculate the transfer function from input V_{in} to output V_{out} .



AutoVSR Framework

Executable Intermediate Representation

format: $V_n N_p [state] N_m value$ | type: independent DC/AC source
gen: $V_{in} 1 s 0 V_{in}$ *# input voltage source with state control*
format: $R_n N_p N_m value$ | passive component
gen: $R1 1 2 R1$ *# resistor between node 1 and node 2*
format: $C_n N_p N_m value$ | energy storage element
gen: $C1 2 0 C1$ *# capacitor connected to ground*
format: $E_n N_o 0 opamp N_p N_m$ | ideal operational amplifier
gen: $E1 3 0 opamp 2 0$ *# op-amp with positive input at node 2*
format: $R_n N_p N_m value$ | feedback resistor
gen: $R2 3 4 R$ *# placeholder resistor (degenerate case)*



Symbolic
Solver
Agent

Validated IR

Step 1: Calculate the transfer function from V_1 to capacitor $C1$

tool: transfer_function_api

Step 2: Summarize and provide final answer

tool: null

Final Answer: $H(s) = \frac{V_{out}}{V_{in}} = -\frac{E_1}{1 + sR_1C_1}$



Performance

